
Modeling Diabetes Risk Using Lifestyle and Demographic Predictors

Natalie Ladocsi^{* 1} Molly O’Leary^{* 1 2} Maya Papadopoulos^{* 3}

Abstract

1. Data:

The data used in this project are from the 2024 National Health Interview Survey (NHIS). This survey is conducted annually among US households and is used to monitor national health trends, as well as inform public health policy. This survey collects data on a broad range of health-related information and uses a large representative sample of the US population. Basic demographic characteristics include height, weight, age group, and BMI. Smoking, drinking, and exercise habits are some of the behavioral characteristics that will be explored later in this project. Using survey data does come with limitations, there may be response or sampling bias. The National Health Interview Survey, since 1957, has conducted confidential face-to-face interviews throughout the year, which may help reduce response bias. Dividing the country into different geographic regions ensures a broad geographic coverage, and randomization will help obtain data from a sample representative of the true population.

The first step with any dataset, especially one this large, is to identify the key variables relating to the initial question. As diabetes is the primary focus of this project, variables related to factors that increase diabetes risk were highlighted. There are three main categories of diabetes found in the dataset: gestational, pre-diabetes, and type I and II. In the original dataset, these three variables are separate [GESDIB_A, PREDIB_A, DIBEV_A]. To increase readability and facilitate drawing conclusions, the three variables will be combined into a single variable. This new variable will take numeric observations, with 1 indicating ‘yes’, 2 in-

dicating ‘no’, 7 indicating ‘refused’, 8 indicating ‘not ascertained’, and 9 indicating ‘don’t know’. Distinguishing between type I and type II diabetes will be included in another variable (DIBTYPE_A) in the answer to the previous variable is 1. In addition, the age of the surveyor when they were first diagnosed with diabetes will have its own variable DIBAGETC_A.

The variables outlined above gather information about the surveyor’s diagnosis of diabetes. Since the goal is also to examine contributing factors to mitigate the risk of diabetes, the predictor variables must be included. One broad category is food insecurity, which contains the variables: [FDSRUNOUT_A, worry food would run out, FDSLAST_A, food didn’t last, FDSHUNGRY_A, hungry because not enough money for food]. Further analysis may reveal which of the variables among the three should be prioritized, or if all are relevant. Additionally, demographic characteristics will be analyzed. These include sex (SEX_A), home location (REGION), race/ethnicity (RACEALLP_A), income level (RATCAT_A), education level of sample adults (EDUCP_A). Lastly, physical lifestyle habits will also be factored through physical activity (MONDNR_A), and weekly walking (WLKEISDAY_A).

The data used were obtained from a survey; therefore, not all surveyors answered every question. In addition, those surveyed may not have been entirely truthful or provided accurate information. This leaves us with NA values that will need to be wrangled before we begin data analysis. Therefore, conclusions can be drawn but cannot be claimed with absolute certainty. This data set has hundreds of variables, so we will remove many variables that we believe to be insignificant. We need to do testing to ensure that we remove the correct ones.

1.1. Pre-Analysis Plan

As outlined in our Data Description, the goal is to develop a model that can classify whether or not a person has diabetes in the United States, based on demographic and behavioral information. Within the data set, information about numerous aspects of the surveyor is outlined, including whether or not they have diabetes. Through training the model and es-

^{*}Equal contribution ¹Department of XXX, University of YYY, Location, Country ²Company Name, Location, Country ³**AUTHORERR: Missing \icmlaffiliation.** . Correspondence to: Firstname1 Lastname1 <first1.last1@xxx.edu>, Firstname2 Lastname2 <first2.last2@www.uk>.

establishing correlation patterns, the goal is to create a model that can estimate whether or not one is likely to develop diabetes depending on the information entered. We will perform different analysis methods such as EDA and PCA to select the variables included in our final model.

After cleaning, organizing, and wrangling the data, we will begin with exploratory data analysis (EDA) to help us better understand the relationships between each variable and the response before modeling. In this step, we will cross-tabulate diabetes with several categorical variables to identify relationships and patterns. Furthermore, the use of bar plots will help us visualize distributions, identify outliers, and show us variables that potentially need scaling. This initial exploratory step will help us select significant variables to include in our next modeling phase. Because the dataset contains many correlated features, we will apply principal component analysis (PCA). The normalized eigenvalues can be plotted with a scree plot, or summed to determine the cumulative R^2 , helping us capture sources of variance in the data.

Once the variables of our analysis have been explored, we plan to use a decision tree model for further analysis. Due to the complexity of our data and the multitude of factors influencing the diabetic outcome of the patient, we felt this was the most appropriate model to use. A decision tree will be compatible with the lack of quantitative or numeric data in our dataset. Trees are also non-reliant on linear relationships which we are not likely to see in our data. The tree itself will provide an easily interpretable visualization of the decision-making pathway. However, since decision trees output a categorical variable, we will have to encode or determine some threshold with diabetes being a yes or no.

In order to accomplish this, we first have to encode the categorical variables using one-hot encoding or `'pd.get_dummies()'`. The tree model cannot handle text labels directly so converting these labels into a numeric form allows the model to run successfully. Next, it is integral to split the data into train and test sets to give the model the ability to generalize. Allowing performance not just on the given data, but predicting for unseen, future data. We will split the data 80/20 into train and test sets, respectively, training the model. After visualization, we will choose to check the accuracy of the model using the confusion matrix method, which depicts how predictions made by the model match the observed outcomes. From there R^2 and RMSE values will be calculated to evaluate the model and determine if changes need to be implemented. One of the biggest concerns with the decision tree model is overfitting which can occur if the model is split too many times. Especially with complex data, we will most likely have to prune the tree to generalize the model.

Model	True Positive Rate	True Negative Rate	R^2
Decision Tree 1	0.82	0.20	0.458
Decision Tree 2	0.842	0.455	0.459

Table 1. Insert Title

2. Methods

2.1. Exploratory Data Analysis (EDA)

Exploratory data analysis was conducted to identify patterns and relationships between predictors and diabetes status. Cross-tabulations and bar plots were used for categorical variables. Boxplots and histograms were used to visualize continuous variables. Correlation analysis showed moderate associations among age, BMI, hypertension, cholesterol, and diabetes status. This step guided feature selection and informed modeling decisions.

2.2. Dimensionality Reduction

Principal component analysis (PCA) was applied to assess multicollinearity and ensure predictors were not linearly dependent on one another.

3. Results

3.1. Diabetes Prevalence and Exploratory Analysis

Initial exploration of the NHIS dataset shows that diabetes prevalence is consistent with population-level estimates. Count plots revealed that a majority of respondents did not report diabetes ($DIAB_ANY = 0$), while roughly 12-15% reported a form of diabetes. Stratification by sex indicated that diabetes is more prevalent among males than it is for females. Correlation analysis of key continuous variables (age, BMI, hypertension, cholesterol) showed moderate associations with diabetes status, consistent with known risk factors.

3.2. Decision Tree Models

Two individual decision tree regressors were trained on the same 80/20 train/test data to predict diabetes status.

The Table 1 results show that the single decision trees are sensitive to training data. The visuals of the trees show them to be deep, meaning that the models are overfitting. True positive rates (TPR) were relatively high, indicating the model predicts diabetic cases well, but the true negative rates (TNR) were varied. This variance suggests a lack of reliability when the model predicts non-diabetic cases. The R^2 values were moderate, the model was able to explain 46% of the variance in diabetes status.

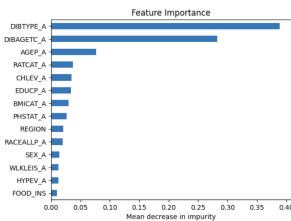


Figure 1. Feature Importance Graph

3.3. Bootstrap Aggregation (Bagging)

To reduce variability and improve model stability, bootstrap aggregation was applied with 1000 resampled decision trees. The ensemble approach produced a more stable estimate, $R^2=0.478$. Compared to single decision trees, the bagged model slightly improved predictive accuracy and reduced the sensitivity to individual training sets. The ensemble methods enhance the robustness of tree-based predictions.

3.4. Initial Analysis

- Overfitting in single trees: The variation in true negative rates across the two single tree models illustrates overfitting, as the trees capture noise in the training data instead of generalized patterns. - Improved stability with bagging: Bootstrap aggregation stabilized predictions and slightly increased R^2 , demonstrating the benefit of ensemble methods for models with high variance. - Feature Importance: Figure 1. illustrates which variables are going to be the most useful when constructing a random forest. Aside from the diabetes variables, the chart shows age to be the most useful among the predictors. The age variable (AGEP_A) has the highest mean decrease in impurity.

ICML uses double-blind review, so author information must not appear. If you are using $\text{\LaTeX}$ and the `icml2025.sty` file, use `\icmlauthor{...}` to specify authors and `\icmlaffiliation{...}` to specify affiliations. (Read the TeX code used to produce this document for an example usage.) The author information will not be printed unless `accepted` is passed as an argument to the style file. Submissions that include the author information will not be reviewed.

3.4.1. SELF-CITATIONS

If you are citing published papers for which you are an author, refer to yourself in the third person. In particular, do not use phrases that reveal your identity (e.g., “in previous work (Langley, 2000), we have shown ...”).

Do not anonymize citations in the reference section. The only exception are manuscripts that are not yet published (e.g., under submission). If you choose to refer to such unpublished manuscripts (Author, 2021), anonymized copies

have to be submitted as Supplementary Material via Open-Review. However, keep in mind that an ICML paper should be self contained and should contain sufficient detail for the reviewers to evaluate the work. In particular, reviewers are not required to look at the Supplementary Material when writing their review (they are not required to look at more than the first 8 pages of the submitted document).

3.4.2. CAMERA-READY AUTHOR INFORMATION

If a paper is accepted, a final camera-ready copy must be prepared. For camera-ready papers, author information should start 0.3 inches below the bottom rule surrounding the title. The authors’ names should appear in 10 point bold type, in a row, separated by white space, and centered. Author names should not be broken across lines. Unbolded superscripted numbers, starting 1, should be used to refer to affiliations.

Affiliations should be numbered in the order of appearance. A single footnote block of text should be used to list all the affiliations. (Academic affiliations should list Department, University, City, State/Region, Country. Similarly for industrial affiliations.)

Each distinct affiliations should be listed once. If an author has multiple affiliations, multiple superscripts should be placed after the name, separated by thin spaces. If the authors would like to highlight equal contribution by multiple first authors, those authors should have an asterisk placed after their name in superscript, and the term “*Equal contribution” should be placed in the footnote block ahead of the list of affiliations. A list of corresponding authors and their emails (in the format Full Name <email@domain.com>) can follow the list of affiliations. Ideally only one or two names should be listed.

A sample file with author names is included in the ICML2025 style file package. Turn on the `[accepted]` option to the stylefile to see the names rendered. All of the guidelines above are implemented by the $\text{\LaTeX}$ style file.

3.5. Abstract

The paper abstract should begin in the left column, 0.4 inches below the final address. The heading ‘Abstract’ should be centered, bold, and in 11 point type. The abstract body should use 10 point type, with a vertical spacing of 11 points, and should be indented 0.25 inches more than normal on left-hand and right-hand margins. Insert 0.4 inches of blank space after the body. Keep your abstract brief and self-contained, limiting it to one paragraph and roughly 4–6 sentences. Gross violations will require correction at the camera-ready phase.

3.6. Partitioning the Text

You should organize your paper into sections and paragraphs to help readers place a structure on the material and understand its contributions.

3.6.1. SECTIONS AND SUBSECTIONS

Section headings should be numbered, flush left, and set in 11 pt bold type with the content words capitalized. Leave 0.25 inches of space before the heading and 0.15 inches after the heading.

Similarly, subsection headings should be numbered, flush left, and set in 10 pt bold type with the content words capitalized. Leave 0.2 inches of space before the heading and 0.13 inches afterward.

Finally, subsubsection headings should be numbered, flush left, and set in 10 pt small caps with the content words capitalized. Leave 0.18 inches of space before the heading and 0.1 inches after the heading.

Please use no more than three levels of headings.

3.6.2. PARAGRAPHS AND FOOTNOTES

Within each section or subsection, you should further partition the paper into paragraphs. Do not indent the first line of a given paragraph, but insert a blank line between succeeding ones.

You can use footnotes¹ to provide readers with additional information about a topic without interrupting the flow of the paper. Indicate footnotes with a number in the text where the point is most relevant. Place the footnote in 9 point type at the bottom of the column in which it appears. Precede the first footnote in a column with a horizontal rule of 0.8 inches.²

3.7. Figures

You may want to include figures in the paper to illustrate your approach and results. Such artwork should be centered, legible, and separated from the text. Lines should be dark and at least 0.5 points thick for purposes of reproduction, and text should not appear on a gray background.

Label all distinct components of each figure. If the figure takes the form of a graph, then give a name for each axis and include a legend that briefly describes each curve. Do not include a title inside the figure; instead, the caption should serve this function.

¹Footnotes should be complete sentences.

²Multiple footnotes can appear in each column, in the same order as they appear in the text, but spread them across columns and pages if possible.

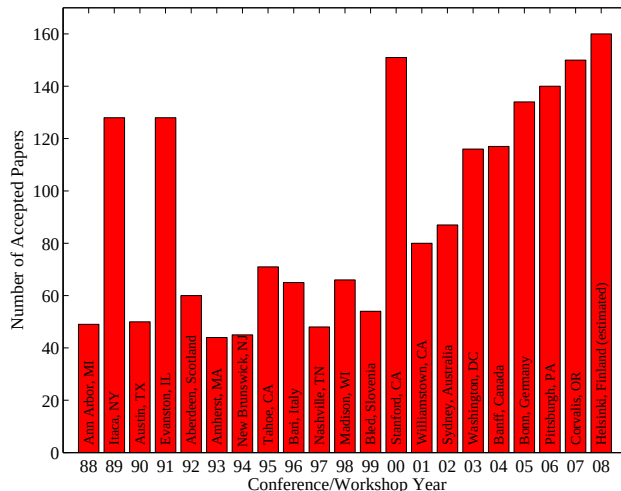


Figure 2. Historical locations and number of accepted papers for International Machine Learning Conferences (ICML 1993 – ICML 2008) and International Workshops on Machine Learning (ML 1988 – ML 1992). At the time this figure was produced, the number of accepted papers for ICML 2008 was unknown and instead estimated.

Algorithm 1 Bubble Sort

Input: data x_i , size m
repeat
 Initialize $noChange = true$.
 for $i = 1$ **to** $m - 1$ **do**
 if $x_i > x_{i+1}$ **then**
 Swap x_i and x_{i+1}
 $noChange = false$
 end if
 end for
until $noChange$ is $true$

Number figures sequentially, placing the figure number and caption *after* the graphics, with at least 0.1 inches of space before the caption and 0.1 inches after it, as in Figure 2. The figure caption should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left. You may float figures to the top or bottom of a column, and you may set wide figures across both columns (use the environment `figure*` in $\LaTeX$). Always place two-column figures at the top or bottom of the page.

3.8. Algorithms

If you are using $\LaTeX$, please use the “algorithm” and “algorithmic” environments to format pseudocode. These require the corresponding stylefiles, `algorithm.sty` and `algorithmic.sty`, which are supplied with this package. Algorithm 1 shows an example.

Table 2. Classification accuracies for naive Bayes and flexible Bayes on various data sets.

DATA SET	NAIVE	FLEXIBLE	BETTER?
BREAST	95.9± 0.2	96.7± 0.2	✓
CLEVELAND	83.3± 0.6	80.0± 0.6	×
GLASS2	61.9± 1.4	83.8± 0.7	✓
CREDIT	74.8± 0.5	78.3± 0.6	
HORSE	73.3± 0.9	69.7± 1.0	×
META	67.1± 0.6	76.5± 0.5	✓
PIMA	75.1± 0.6	73.9± 0.5	
VEHICLE	44.9± 0.6	61.5± 0.4	✓

3.9. Tables

You may also want to include tables that summarize material. Like figures, these should be centered, legible, and numbered consecutively. However, place the title *above* the table with at least 0.1 inches of space before the title and the same after it, as in Table 2. The table title should be set in 9 point type and centered unless it runs two or more lines, in which case it should be flush left.

Tables contain textual material, whereas figures contain graphical material. Specify the contents of each row and column in the table’s topmost row. Again, you may float tables to a column’s top or bottom, and set wide tables across both columns. Place two-column tables at the top or bottom of the page.

3.10. Theorems and such

The preferred way is to number definitions, propositions, lemmas, etc. consecutively, within sections, as shown below.

Definition 3.1. A function $f : X \rightarrow Y$ is injective if for any $x, y \in X$ different, $f(x) \neq f(y)$.

Using Definition 3.1 we immediately get the following result:

Proposition 3.2. *If f is injective mapping a set X to another set Y , the cardinality of Y is at least as large as that of X*

Proof. Left as an exercise to the reader. $\square$

Lemma 3.3 stated next will prove to be useful.

Lemma 3.3. *For any $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ injective functions, $f \circ g$ is injective.*

Theorem 3.4. *If $f : X \rightarrow Y$ is bijective, the cardinality of X and Y are the same.*

An easy corollary of Theorem 3.4 is the following:

Corollary 3.5. *If $f : X \rightarrow Y$ is bijective, the cardinality of X is at least as large as that of Y .*

Assumption 3.6. The set X is finite.

Remark 3.7. According to some, it is only the finite case (cf. Assumption 3.6) that is interesting.

3.11. Citations and References

Please use APA reference format regardless of your formatter or word processor. If you rely on the $\text{\LaTeX}$ bibliographic facility, use `natbib.sty` and `icml2025.bst` included in the style-file package to obtain this format.

Citations within the text should include the authors’ last names and year. If the authors’ names are included in the sentence, place only the year in parentheses, for example when referencing Arthur Samuel’s pioneering work (1959). Otherwise place the entire reference in parentheses with the authors and year separated by a comma (Samuel, 1959). List multiple references separated by semicolons (Kearns, 1989; Samuel, 1959; Mitchell, 1980). Use the ‘et al.’ construct only for citations with three or more authors or after listing all authors to a publication in an earlier reference (Michalski et al., 1983).

Authors should cite their own work in the third person in the initial version of their paper submitted for blind review. Please refer to Section 3.2 for detailed instructions on how to cite your own papers.

Use an unnumbered first-level section heading for the references, and use a hanging indent style, with the first line of the reference flush against the left margin and subsequent lines indented by 10 points. The references at the end of this document give examples for journal articles (Samuel, 1959), conference publications (Langley, 2000), book chapters (Newell & Rosenbloom, 1981), books (Duda et al., 2000), edited volumes (Michalski et al., 1983), technical reports (Mitchell, 1980), and dissertations (Kearns, 1989).

Alphabetize references by the surnames of the first authors, with single author entries preceding multiple author entries. Order references for the same authors by year of publication, with the earliest first. Make sure that each reference includes all relevant information (e.g., page numbers).

Please put some effort into making references complete, presentable, and consistent, e.g. use the actual current name of authors. If using `bibtex`, please protect capital letters of names and abbreviations in titles, for example, use `\B{ayesian}` or `\L{ipschitz}` in your `.bib` file.

Results

Authors are kindly asked to make their submissions as accessible as possible for everyone including people with disabilities and sensory or neurological differences. Tips of how to achieve this and what to pay attention to will be provided on the conference website <http://icml.cc/>.

Software and Data

If a paper is accepted, we strongly encourage the publication of software and data with the camera-ready version of the paper whenever appropriate. This can be done by including a URL in the camera-ready copy. However, **do not** include URLs that reveal your institution or identity in your submission for review. Instead, provide an anonymous URL or upload the material as “Supplementary Material” into the OpenReview reviewing system. Note that reviewers are not required to look at this material when writing their review.

Acknowledgements

Do not include acknowledgements in the initial version of the paper submitted for blind review.

If a paper is accepted, the final camera-ready version can (and usually should) include acknowledgements. Such acknowledgements should be placed at the end of the section, in an unnumbered section that does not count towards the paper page limit. Typically, this will include thanks to reviewers who gave useful comments, to colleagues who contributed to the ideas, and to funding agencies and corporate sponsors that provided financial support.

Impact Statement

Authors are **required** to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered section at the end of the paper (co-located with Acknowledgements – the two may appear in either order, but both must be before References), and does not count toward the paper page limit. In many cases, where the ethical impacts and expected societal implications are those that are well established when advancing the field of Machine Learning, substantial discussion is not required, and a simple statement such as the following will suffice:

“This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.”

The above statement can be used verbatim in such cases, but we encourage authors to think about whether there is content which does warrant further discussion, as this statement will be apparent if the paper is later flagged for ethics review.

References

Author, N. N. Suppressed for anonymity, 2021.

Duda, R. O., Hart, P. E., and Stork, D. G. *Pattern Classification*. John Wiley and Sons, 2nd edition, 2000.

Kearns, M. J. *Computational Complexity of Machine Learning*. PhD thesis, Department of Computer Science, Harvard University, 1989.

Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.

Michalski, R. S., Carbonell, J. G., and Mitchell, T. M. (eds.). *Machine Learning: An Artificial Intelligence Approach, Vol. I*. Tioga, Palo Alto, CA, 1983.

Mitchell, T. M. The need for biases in learning generalizations. Technical report, Computer Science Department, Rutgers University, New Brunswick, MA, 1980.

Newell, A. and Rosenbloom, P. S. Mechanisms of skill acquisition and the law of practice. In Anderson, J. R. (ed.), *Cognitive Skills and Their Acquisition*, chapter 1, pp. 1–51. Lawrence Erlbaum Associates, Inc., Hillsdale, NJ, 1981.

Samuel, A. L. Some studies in machine learning using the game of checkers. *IBM Journal of Research and Development*, 3(3):211–229, 1959.

A. You *can* have an appendix here.

You can have as much text here as you want. The main body must be at most 8 pages long. For the final version, one more page can be added. If you want, you can use an appendix like this one.

The `\onecolumn` command above can be kept in place if you prefer a one-column appendix, or can be removed if you prefer a two-column appendix. Apart from this possible change, the style (font size, spacing, margins, page numbering, etc.) should be kept the same as the main body.